A Low-Cost Miniature Mobile Robotic Platform for Education and Swarm Robotics Experiments

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***Abstract*—This paper presents the design, implementation, and preliminary characterization of a low-cost miniature mobile robotic platform intended for swarm robotics research and STEM education. The proposed system integrates a fully 3D-printed PLA chassis with a modular electronic architecture based on the ESP32 microcontroller, incorporating Wi-Fi/Bluetooth wireless communication, eight infrared proximity sensors, and a differential-drive locomotion system actuated by 28BYJ-48 stepper motors. The platform is designed for ease of fabrication and replication using off-the-shelf components. Key contributions include a UDP-based automatic robot discovery mechanism for multi-robot deployment without manual network configuration, and a cooperative time-driven firmware architecture that coordinates sensing, actuation, and communication without a real-time operating system. A mobility trial covering five consecutive 1 m segments recorded a mean translational speed of 51.20 mm/s (σ = 0.02 mm/s) over a 5 m straight path, confirming consistent and repeatable locomotion.**

***Index Terms*—Swarm Robotics, STEM Education, Low-Cost Robotics, 3D Printing.**

I. Introduction

Educational robotics has been widely adopted as a practical approach to teaching concepts in Science, Technology, Engineering, and Mathematics (STEM), promoting active learning through experimentation and hands-on activities. Beyond supporting the development of problem-solving, creativity, and teamwork skills, educational robotics also provides an accessible environment for introducing advanced topics in robotics. Among these topics, swarm robotics has emerged as an active research field that investigates how large groups of relatively simple robots can achieve complex collective behaviors through local interactions.

Since swarm robotics experiments typically require multiple robots operating simultaneously, platform cost, availability, and reproducibility become critical factors for both research and educational applications. Although platforms such as e-puck [1], Kilobot [2], and Highly Educational Robot (HeRo) [3] are widely used in research and education, scaling their use to larger groups of robots may present practical challenges. For example, the e-puck is priced at approximately $865 USD for Version 2 and $1,050 USD for Version 1, the Kilobot provides limited sensing capabilities, relies on vibration-based locomotion, and has been discontinued, whereas the HeRo does not include an onboard camera. These challenges motivate the development of alternative platforms that are accessible, scalable, and easy to reproduce.

To address these challenges, this work proposes a miniature mobile robotic platform designed for educational and swarm robotics applications. The platform emphasizes modularity, ease of manufacturing, and the use of locally available components to facilitate experimentation in educational and research environments while providing a flexible architecture that can be adapted to different application requirements.

II. Related Work

This section discusses representative multi-robot platforms reported in the literature for educational and research applications. Their main characteristics, architectures, and functionalities are analyzed in order to identify strengths, limitations, and design choices that informed the development of the platform proposed in this work.

e-puck 1. [1] The e-puck is a mobile robotic platform developed at the École Polytechnique Fédérale de Lausanne (EPFL). It has diameter of $70\,\mathrm{mm}$, a height of $55\,\mathrm{mm}$, and a mass of $150\,\mathrm{g}$. The platform is based on a dsPIC 30F6014A microcontroller operating at $60\,\mathrm{MHz}$, with $8\,\mathrm{kB}$ of RAM and $144\,\mathrm{kB}$ of Flash memory [4]. Locomotion is achieved through a differential-drive mechanism with a maximum linear speed of up to $150\,\mathrm{mm/s}$. The platform is powered by a lithium-ion battery of $5\,\mathrm{W\,h}$, providing approximately $3\,\mathrm{h}$ of autonomy. It costs 850 CHF (equivalent to $1,050 USD) [5].

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e-puck 2. [5] builds upon the original e-puck 1 platform maintaining mechanical compatibility. The platform is based on a STM32F4 microcontroller, and includes Wi-Fi, Bluetooth/BLE, and USB, together with an expanded sensor suite that includes a color camera, a time-of-flight (ToF) proximity sensors, an inertial measurement unit (IMU), four microphones, RGB LEDs, and onboard debugging and programming support. It costs 700 CHF (equivalent to $865 USD) [5].

Kilobot. [2] The Kilobot was designed specifically for swarm robotics experiments, enabling large-scale studies with hundreds of units. It has a diameter of $3.3\,\mathrm{cm}$, a height of $3.4\,\mathrm{cm}$, and a mass of approximately $17\,\mathrm{g}$. Locomotion is achieved through two vibration motors, yielding an average speed of approximately $0.04\,\mathrm{km\,h^{-1}}$, equivalent to $11.1\,\mathrm{mm/s}$. Its ATmega328 microcontroller provides $2\,\mathrm{kB}$ of RAM and $32\,\mathrm{kB}$ of Flash memory. The parts to assemble the robot cost around $15 USD per unit, making it well suited for large-scale deployments; however, the platform has since been commercially discontinued.

HeRo. [3] The HeRo is a fully open-source platform compatible with the Robot Operating System (ROS). It employs servo motors to achieve a balance among size, speed, and control, and parts for assembling one robot are priced at around $18 USD.

Jasmine. [6] Developed under the Swarmrobot.org initiative, the Jasmine is an open-hardware platform for large-scale swarm experiments [7]. It features a diameter of $26\,\mathrm{mm}$, a height of $26\,\mathrm{mm}$, and a mass of $30\,\mathrm{g}$. Its ATmega16 microcontroller ($1\,\mathrm{kB}$ RAM, $16\,\mathrm{kB}$ Flash) handles sensing and communication via six IR proximity sensors with a range of up to $15\,\mathrm{cm}$. The parts required to assemble the robot cost approximately $126 USD.

Swarmanoid. [8] The Swarmanoid framework investigates heterogeneous swarm cooperation, demonstrating how simple cooperative agents can collectively solve complex tasks. Its conceptual foundations provide both theoretical and educational motivation for the platform proposed in this work.

Alice. [9] The Alice robot, developed at EPFL, is a miniature platform measuring roughly $2\,\mathrm{cm} \times 2\,\mathrm{cm} \times 2\,\mathrm{cm}$. It supports autonomous movement, environmental sensing, and local inter-robot communication, with optional extension modules for enhanced capabilities.

Other platforms. The Thymio [10], Colias [11], GRITSBot [12], mBot [13], and TurtleBot [14] are additional platforms reported in the literature. While each addresses specific research or educational goals, they present trade-offs in cost, scalability, or hardware flexibility that may limit their use in large-scale or resource-constrained scenarios.

Table I summarizes the main characteristics of the miniature mobile robotic platforms discussed in this section. The comparison focuses on physical dimensions, cost, and battery autonomy, highlighting the design trade-offs among the different platforms and positioning the proposed platform with respect to existing alternatives.

A more detailed comparison of the sensing, communication, and locomotion capabilities of these platforms is presented in Table II. The comparison highlights the availability of key hardware components, including inertial sensors, infrared proximity sensing, wireless communication interfaces, and locomotion mechanisms.

TABLE I: Comparative overview of miniature mobile robotic platforms. All values are approximate. The cost of the proposed robot is based on bulk purchasing and excludes freight and import tax, while the cost of the other platforms are those reported in their respective original publications.

Robot	Size (mm)	Cost (USD)	Autonomy (h)
e-puck 1	∅70 × 55	1050	3
e-puck 2	∅70 × 55	865	3
Kilobot	∅33 × 34	15*	3–24
HeRo	30–40 (3D)	18*	3
Jasmine	∅26 × 26	126*	N/A
Swarmanoid	380 × 440 × 300	N/A	10–20
Alice	20 × 20 × 20	300 – 500	8–10
Proposed	∅76 × 83**	35 – 50*	>3

* Parts only. ** Height measured up to the PCB surface. **N/A**: not reported.

The platforms reviewed above reveal a consistent trade-off between sensing capabilities and scalability. Feature-rich platforms such as the e-puck offer extensive sensing and communication capabilities but have unit costs exceeding $850 USD, making large-scale deployments economically impractical for educational settings.

In contrast, minimalist platforms such as the Kilobot and Jasmine reduce cost at the expense of locomotion precision and sensing diversity. None of the platforms reviewed in this work combines wireless zero-configuration deployment, camera sensing, infrared obstacle detection, and stepper-motor-based differential drive within within a reproducible open-hardware design with a total hardware cost below $50 USD.

III. Proposed Platform

The proposed platform is a miniature mobile robot designed for educational and swarm robotics applications. It features a modular mechanical and electronic architecture integrating wireless communication, eight RGB LEDS, a camera, inertial and IR-based proximity sensing, and differential-drive locomotion with two independent wheels. The design emphasizes ease of fabrication, maintenance, and replication using widely available components, making it suitable for multi-robot experimentation.

A. Mechanical Design

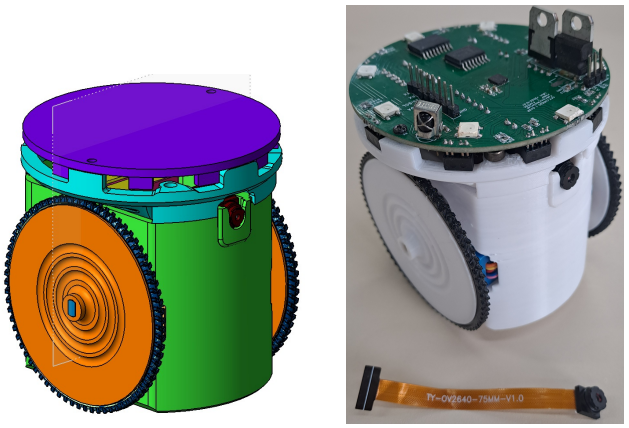
The robot structure is illustrated in Figure 1. The CAD model shown in Figure 1a presents the main mechanical components, including the chassis (green), the Printed Circuit Board (PCB) serving as the control board (purple), the PCB support ring (cyan), and the lateral wheels (orange). This layered design facilitates assembly, maintenance, and access to the electronic components while preserving a compact mechanical structure.

TABLE II: Component selection across robotic platforms.

Component	e-puck 2	Kilobot	HeRo	Jasmine	Swarmanoid	Alice	Proposed
IR Sensor	✓	✓	✓	✓	✓	✓	✓
Ultrasonic	✓		✓		✓	✓	
Accelerometer	✓		✓		✓	✓	✓
Gyroscope	✓		✓		✓	✓	✓
WiFi	✓		✓		✓	✓	✓
Bluetooth	✓		✓				✓
IR Communication	✓	✓	✓	✓	✓	✓	✓
Camera	✓				✓	✓	✓
Wheel-based Locomotion	✓		✓	✓	✓	✓	✓
Vibration-based Locomotion		✓					
LEDs	✓	✓	✓	✓	✓	✓	✓
Sound	✓		✓		✓	✓	
Number of supported features	11	4	10	4	10	10	9

Check marks (✓) indicate that the corresponding component is available on the robotic platform.
Blank cells indicate that the component is not available.

All structural components were fabricated using Fused Deposition Modeling (FDM) 3D printing with PLA thermoplastic filament. The resulting prototype, shown in Figure 1b, features an exposed PCB on the top, with the electronic components mounted on its upper and lower surfaces, while the stepper motors, batteries, and ESP32 microcontroller are enclosed within the structure beneath the PCB.

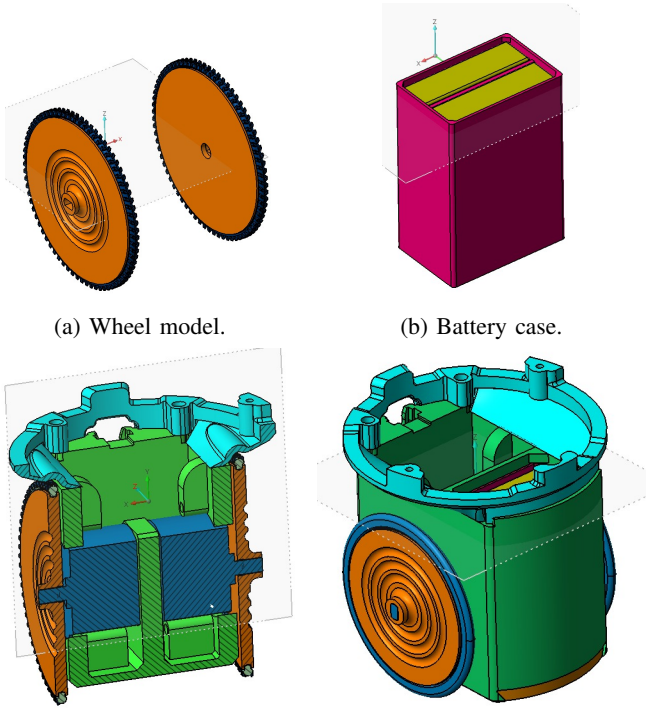


(a) CAD model of the cylindrical enclosure, view from the front.

(b) FDM 3D-printed physical prototype.

Fig. 1: Robotic platform casing.

Additional details of the mechanical design are presented in Figure 2. The wheels, shown in Figure 2a, are directly mounted on the stepper motor shafts (shown in blue in Figure 2c), with a 3D-printed toothed belt fitted around their circumference to provide traction. Figure 2b illustrates the removable battery compartment, located at the rear and accessible from below, which allows the two-cell battery pack to be replaced without disassembling the main chassis. A cross-sectional view of the robot is provided in Figure 2c, highlighting the internal structural layout and the arrangement of the mechanical components. Figure 2d highlights the motor compartment toward the front and the battery compartment at the rear.



(a) Wheel model.

(b) Battery case.

(c) Cross-sectional view from the rear.

(d) View from the rear.

Fig. 2: Three-dimensional CAD models.

B. Electronic Architecture

The electronic architecture of the proposed platform comprises the main subsystems responsible for sensing, processing, actuation, and communication. Figure 3 illustrates the connection between the ESP-WROVER-DEV microcontroller board and the robot controller board. This interface enables

continuous data acquisition from the sensors and peripherals connected to the platform.

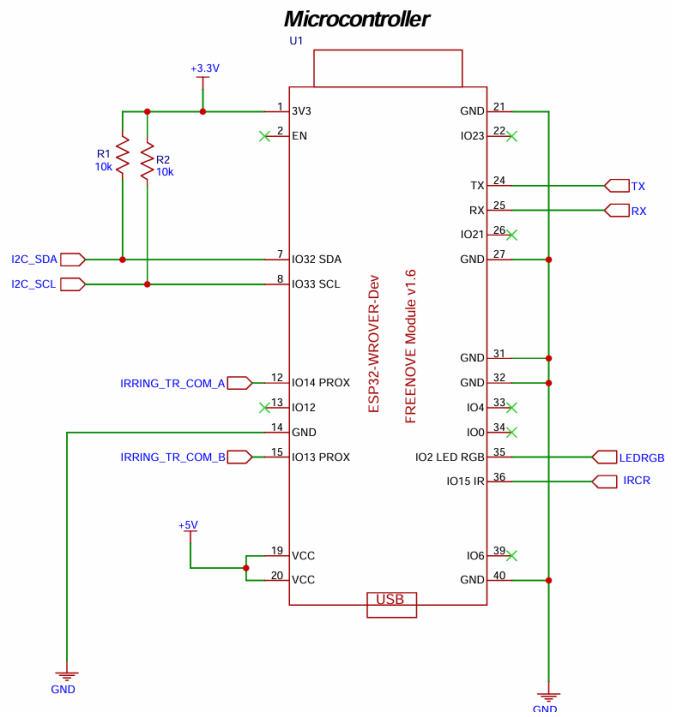


Fig. 3: Schematic of the microcontroller board connection.

The Inertial Measurement Unit (IMU), responsible for acceleration and angular rate sensing, is connected via I2C. The IMU integrates a 3-axis accelerometer and a 3-axis gyroscope on a single chip, enabling the acquisition and transmission of motion-related data such as linear acceleration and angular velocity.

Figure 4 illustrates the control circuit for the 28BYJ-48 stepper motors. The motors are driven through a ULN2803AG driver, which receives eight control signals from the PCF8574 I2C I/O expander. The circuit operates from a +5 V supply, with the ULN2803AG acting as a driver stage between the controller and the motors.

The system also includes eight short-range infrared sensors (TCRT 1000) for obstacle detection, each composed of an IR LED emitter and receiver, as shown in Figure 5. The infrared sensors are read in four cycles, with two most distant pairs activated simultaneously in each cycle, such that all sensors are sampled while avoiding interference between neighboring sensors. This activation scheme is controlled via the I2C protocol by a PCF8574 I2C I/O expander. A dual 4-to-1 analog multiplexer multiplexes the eight sensor readings onto two ESP32 input pins.

The battery and voltage regulation circuit ensures a stable +5 V supply for all components, with a dedicated power rail for the motors (+5VM), as shown in Figure 6. The power system is based on two rechargeable LiPo (Lithium Polymer) batteries, as shown in Figure 2b. Each battery provides a

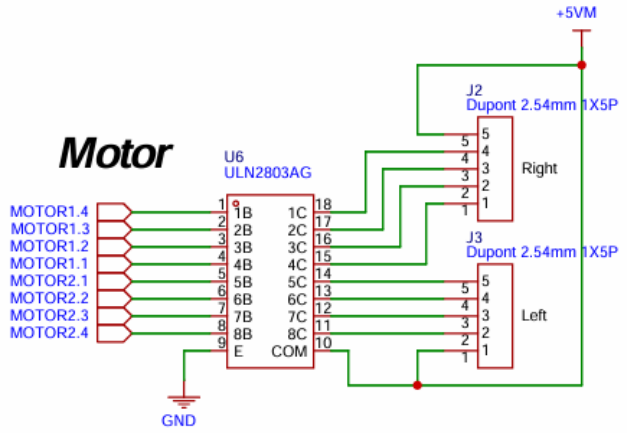


Fig. 4: Stepper motor driver circuit based on the ULN2803AG.

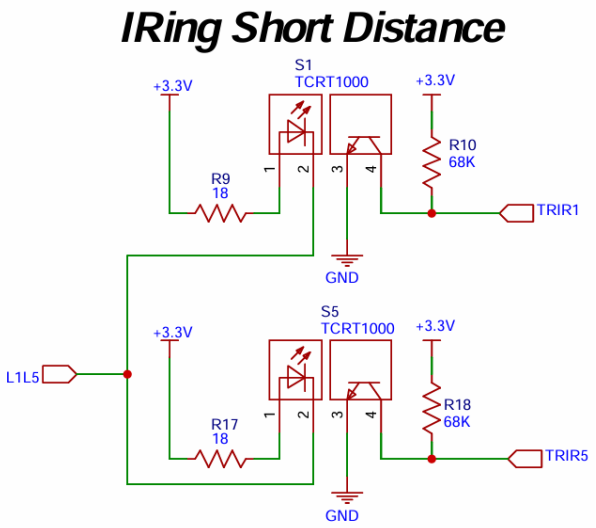


Fig. 5: Schematic of one of the four pairs of infrared (IR) sensors; only two of the eight sensors are shown. Each TCRT1000 integrates an IR LED emitter and a phototransistor receiver.

nominal voltage of 3.7 V, and their series configuration results in up to 7.4 V, decreasing as the batteries discharge.

The electronic circuit board integrates the main control and interfacing components of the platform. Figure 7a presents the top view, showing the switch and RGB LEDs. Figure 7b shows the bottom view of the assembly, where the ESP-WROVER-32 module is coupled to the custom controller board.

C. Communication and Programming

Wireless communication is a fundamental requirement of the proposed robotic platform, enabling both remote programming and real-time monitoring without requiring physical intervention on individual units. To support these functionalities, the ESP32 microcontroller integrates a dual-mode wireless interface with Wi-Fi connectivity, which is used to establish a bidirectional communication channel between the robot and

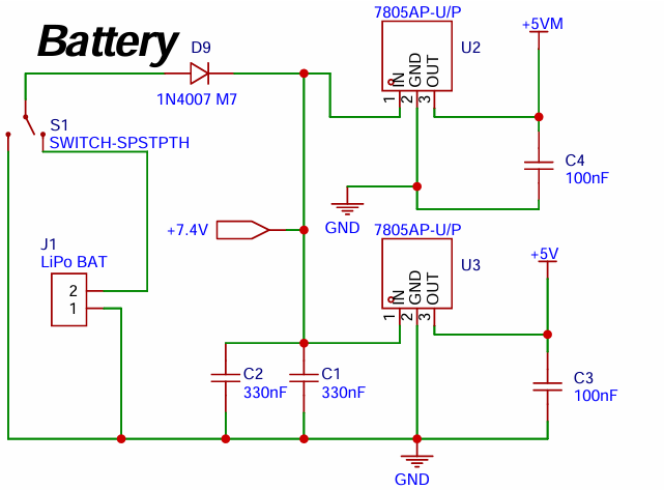
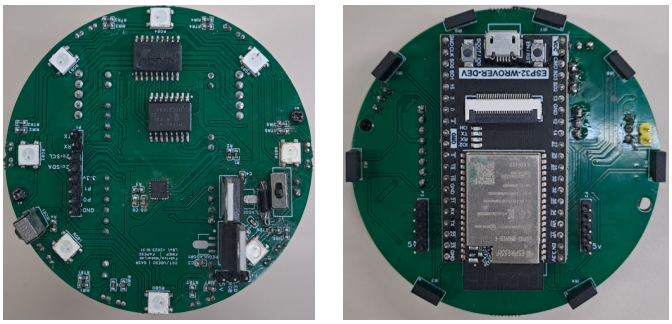


Fig. 6: Battery power supply circuit schematic.



(a) Top view of the control board assembly.

(b) Bottom view of the control board assembly.

Fig. 7: Electronic circuit board assembly.

a host infrastructure. The firmware implements a lightweight HTTP server running directly on the microcontroller, exposing a REST-based interface through which the robot identity, operational status, and sensor readings can be queried.

During initialization, each robot broadcasts its presence over the local network through a UDP-based discovery service, allowing a central management system to automatically identify active units without prior knowledge of their network addresses. This mechanism is particularly advantageous in multi-robot deployments, where the simultaneous operation of several robots would otherwise require manual network configuration and address assignment.

When no valid Wi-Fi configuration is available, the firmware automatically switches to Access Point mode and provides a captive portal through which network credentials can be configured using a standard web browser. This approach reduces the dependency on external configuration tools and simplifies the deployment process in educational and laboratory environments.

From a software architecture perspective, the firmware adopts a cooperative scheduling model based on a time-driven event dispatcher, eliminating the need for a real-time operating system. This execution model allocates processing time to low-level peripheral management while maintaining the responsiveness of the network services executed within the same main loop.

IV. Locomotion Characterization

The robot employs a differential-drive locomotion system in which two independently actuated stepper motors drive the left and right wheel assemblies. Velocity commands are represented as signed integer values, where the magnitude defines the stepping frequency and the sign specifies the direction of rotation. These commands are transmitted over an I2C bus to a PCF8574 port expander, which interfaces directly with the motor driver while reducing the GPIO requirements of the main controller.

The mobility of the proposed robotic platform was evaluated through a straight-line motion experiment designed to assess speed consistency and directional stability under controlled indoor conditions. A single continuous 5 m straight-line trial was performed on a flat indoor surface and divided into five consecutive 1 m segments for analysis.

Robot displacement and elapsed time were obtained through frame-by-frame video analysis using the Tracker software (version 6.1), with a calibrated virtual ruler overlaid on the recorded footage to convert pixel coordinates into metric units. Approximately 3,000 timestamped position samples were acquired during the experiment, allowing the time required to traverse each 1 m segment to be determined with high temporal resolution and low measurement noise.

During operation, displacement is estimated through software-based step counting by incrementing or decrementing independent accumulators at each motor phase transition. Although this approach is subject to accumulated error in the absence of external localization, it provides a useful estimate of relative displacement for short-duration navigation tasks and proportional closed-loop control in bounded environments. Additional trials at different step frequencies are planned as future work to further characterize the relationship between step frequency and translational speed.

The robot is powered by two 2400 mAh LiPo batteries connected in series. To evaluate its operational autonomy, the robot was driven continuously while performing an in-place rotation, with the left and right wheels operating at +1000 and -1000 steps/s, respectively, from a fully charged state until either the batteries were depleted or three hours had elapsed.

A. Results

The straight-line mobility experiment showed that the robot maintained a nearly constant velocity throughout the 5 m path, completing the trial in 97.66 s with an average segment time of 19.53 s (sample $\sigma = 0.02$ s), as summarized in Table III. The low standard deviations indicate a highly repeatable locomotion behavior under the evaluated conditions, with the remaining minor variations primarily attributed to odometry discretization and timestamp quantization during spatial segmentation.

TABLE III: Straight-line mobility characterization.

Segment (m)	Segment Time (s)	Elapsed Time (s)	Speed (mm/s)
0–1	19.54	19.54	51.18
1–2	19.54	39.08	51.18
2–3	19.50	58.58	51.28
3–4	19.54	78.12	51.18
4–5	19.54	97.66	51.18
Mean	19.53	–	51.20
Std. Dev.	0.01	–	0.02

The measured mean linear speed was 51.20 mm/s (sample $\sigma = 0.02$ mm/s), indicating consistent locomotion throughout the evaluated trajectory.

The battery discharge behavior observed during six consecutive 30-minute tests is summarized in Table IV. All experiments were performed under constant load at room temperature (≈25 °C), allowing the discharge profile to be compared with the characteristic three-phase behavior of LiPo batteries.

TABLE IV: Autonomy test battery lipo.

Test	Time (min)	Initial (V)	Final (V)	Drop (V)
1	30	8.03	7.52	0.51
2	30	7.51	7.40	0.11
3	30	7.40	7.00	0.4
4	30	7.00	6.90	0.1
5	30	6.90	6.75	0.15
6	30	6.75	6.60	0.15

As expected, the largest voltage drop occurred during Test 1 ($0.51\,\mathrm{V}$), corresponding to the initial discharge phase immediately after a fully charged state. Test 2 exhibited a substantially smaller drop ($0.11\,\mathrm{V}$). A larger decrease was again observed in Test 3 ($0.40\,\mathrm{V}$). During Tests 4 to 6, the voltage drop remained between $0.10\,\mathrm{V}$ to $0.15\,\mathrm{V}$ per interval, indicating a gradual decline toward the recommended minimum operating voltage of $3.0\,\mathrm{V}$ per cell ($6.0\,\mathrm{V}$ for a 2S pack).

Over the entire 180 min evaluation, the battery terminal voltage decreased from $8.03\,\mathrm{V}$ to $6.60\,\mathrm{V}$, an overall reduction of $1.43\,\mathrm{V}$. Since the final voltage remained above the recommended cutoff threshold after 180 min of continuous operation, the results indicate that the proposed platform provides sufficient battery autonomy for at least three hours of operation.

V. Conclusions

This paper presented the design, implementation, and preliminary characterization of a low-cost miniature mobile robotic platform intended for educational and swarm robotics applications. The proposed system integrates a fully 3D-printed modular mechanical structure with an ESP32-based electronic architecture providing wireless communication, an eight-LED RGB ring for visual feedback, infrared proximity detection, and differential-drive locomotion using stepper motors.

These design choices provide low manufacturing cost, ease of fabrication, and functional capability through a modular hardware architecture and an integrated software framework. The hardware design simplifies assembly, maintenance, and future extensions, while the software architecture enables automatic multi-robot deployment through wireless communication and automatic network discovery. Preliminary experimental characterization demonstrated consistent and repeatable mobility together with three hours of continuous battery operation. These characteristics provide an effective balance between affordability, modularity, and functionality, making the proposed platform an accessible and reproducible alternative for educational institutions and research groups.

Future work will include a broader quantitative evaluation of communication performance, different mobility patterns, and long-term mobility, as well as the implementation and validation of swarm robotics behaviors employing the available sensing capabilities across the behaviors evaluated.

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